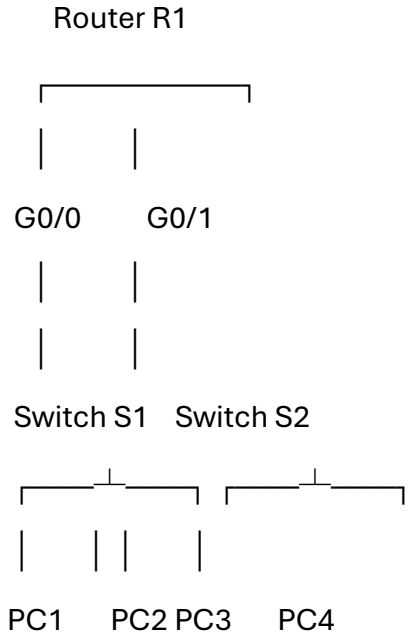
Ex.No.10	Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer
DATE:	

AIM:

To design and configure a simple network topology with one router, two switches, and four PCs in Cisco Packet Tracer for successful inter-network communication.

PROCEDURE:

1. Topology



2. Devices to place

In **Cisco Packet Tracer**, place:

- **1 Router** – Cisco 1941
- **2 Switches** – Cisco 2960
- **4 PCs**

Use **Copper Straight-Through** cables.

3. IP Addressing

Device	Interface	IP Address	Subnet Mask	Gateway
R1	G0/0	192.168.1.1	255.255.255.0	—

Device	Interface	IP Address	Subnet Mask	Gateway
R1	G0/1	192.168.2.1	255.255.255.0	—
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.255.255.0	192.168.2.1

So we have **two different networks**:

Network 1: 192.168.1.0/24

PC1 ─┐

PC2 ─┤

S1 ── R1 G0/0 = 192.168.1.1

Network 2: 192.168.2.0/24

PC3 ─┐

PC4 ─┤

S2 ── R1 G0/1 = 192.168.2.1

4. Configure Router R1

Click **R1 → CLI** and enter:

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

end

write memory

The important command is:

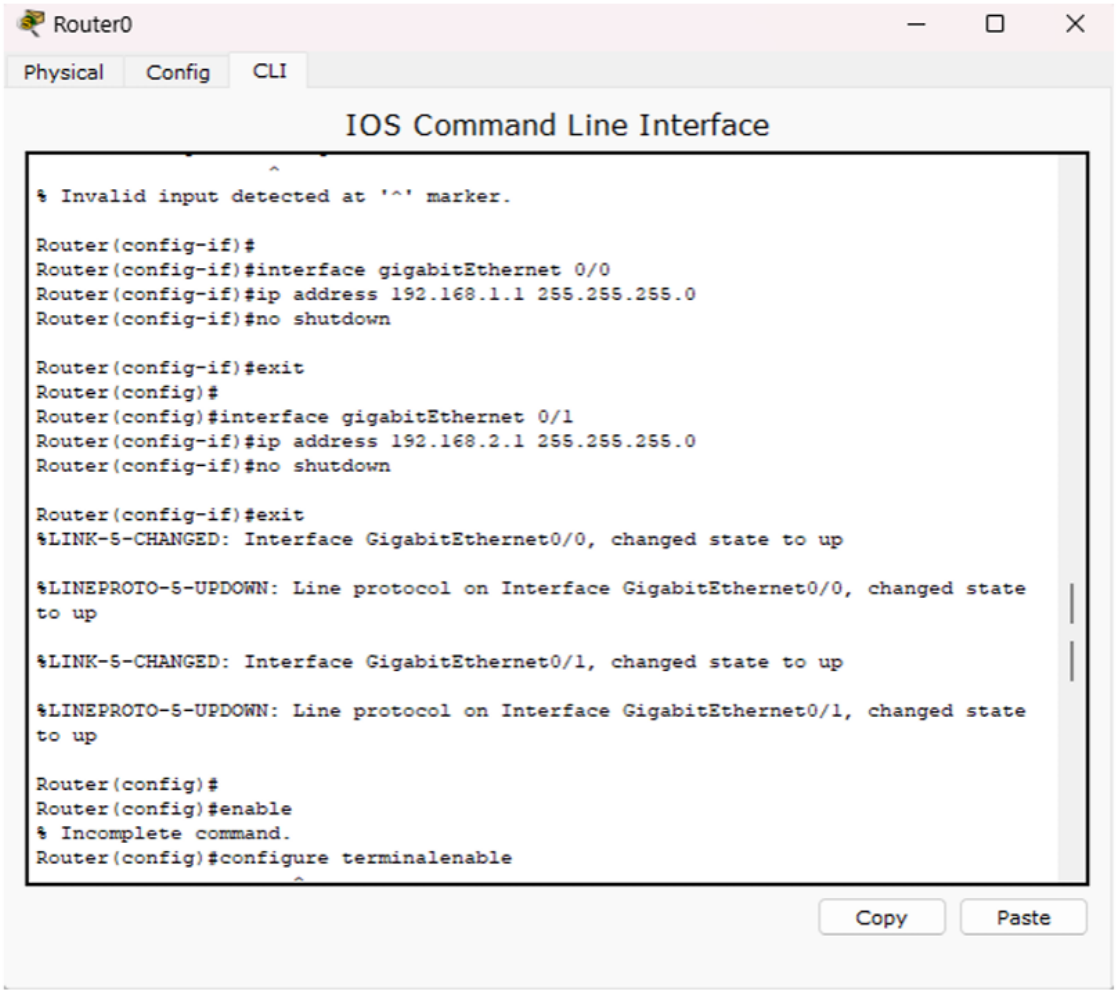
no shutdown

It activates the router interface.

Check the interfaces using:

show ip interface brief

You should see the interfaces as **up/up**.



5. Configure PC1

Go to:

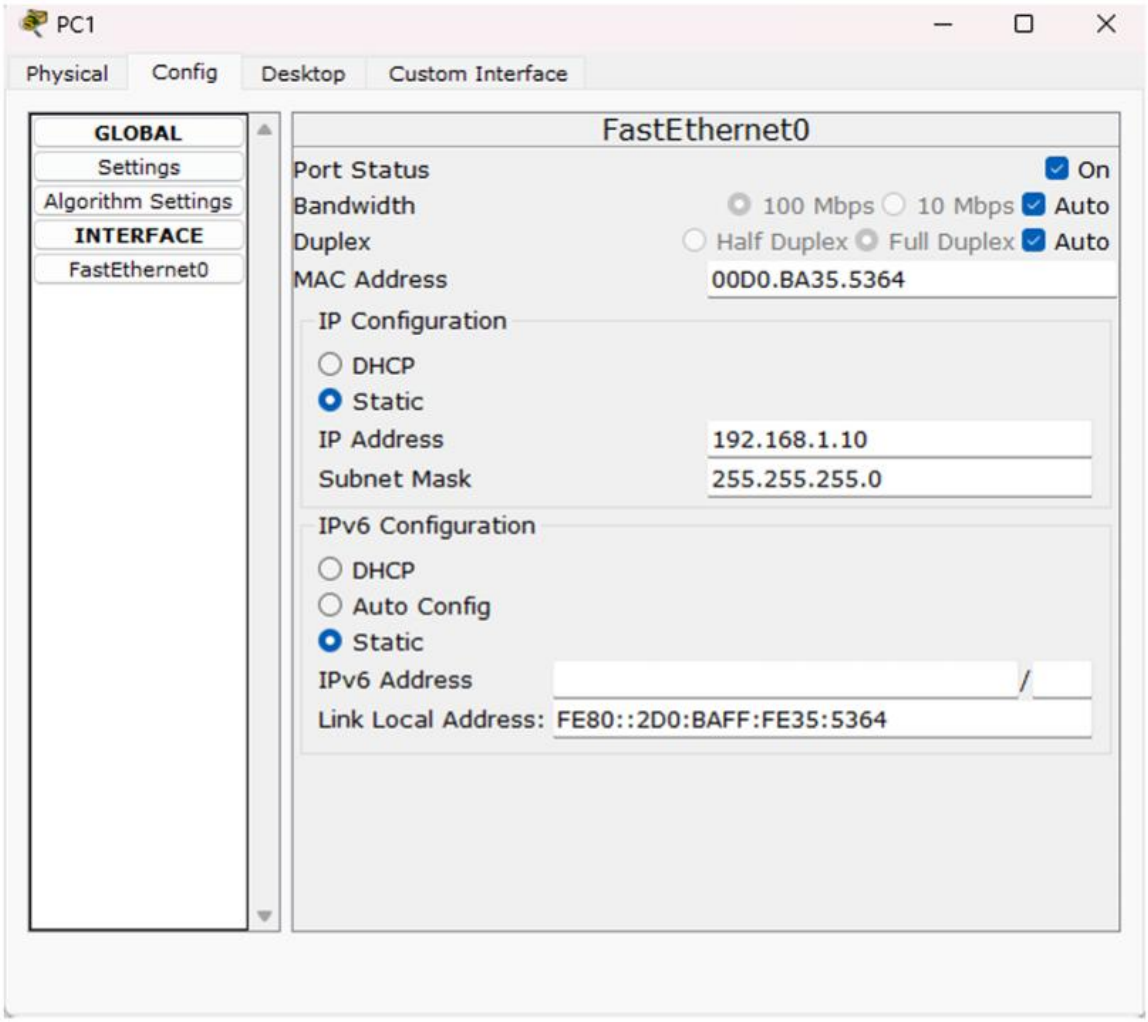
PC1 → Desktop → IP Configuration

Enter:

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

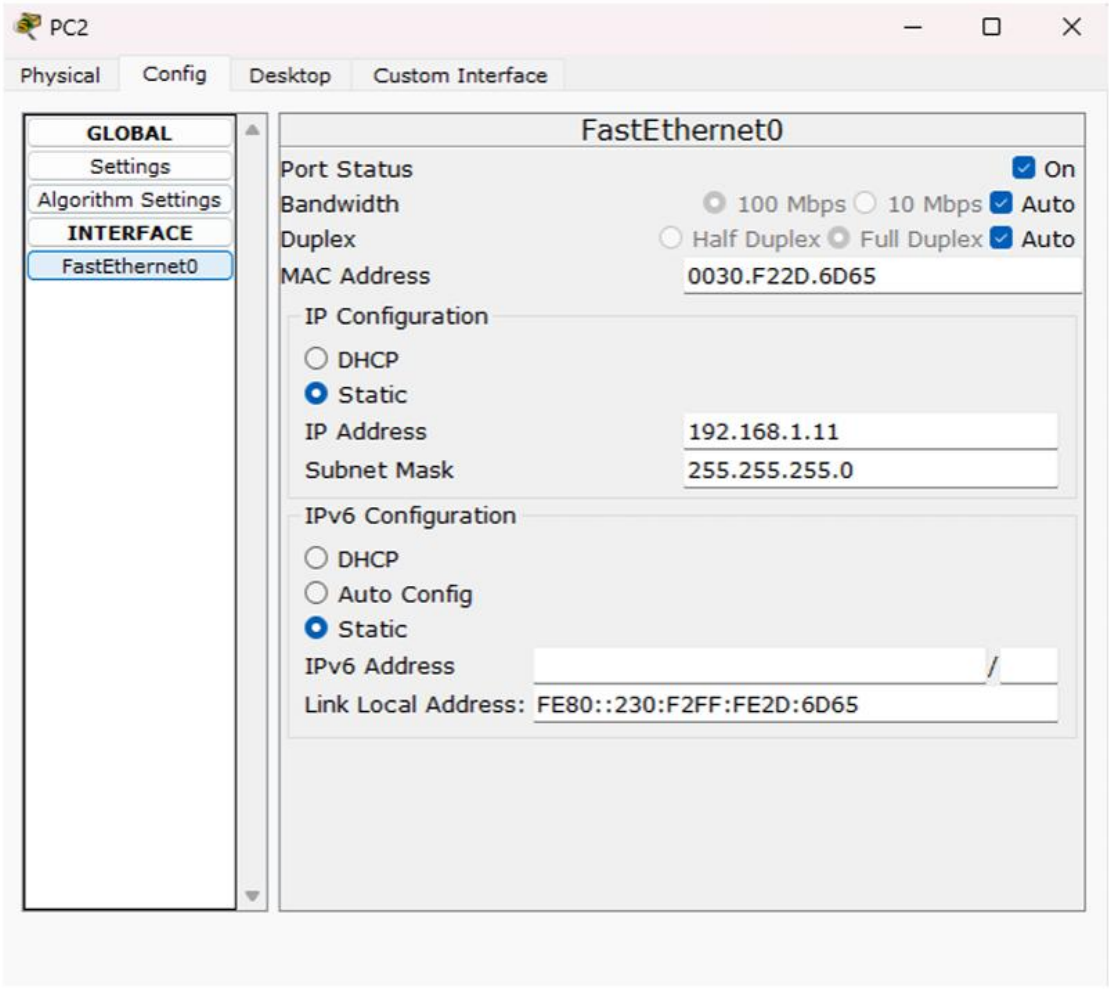


6. Configure PC2

IP Address: 192.168.1.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

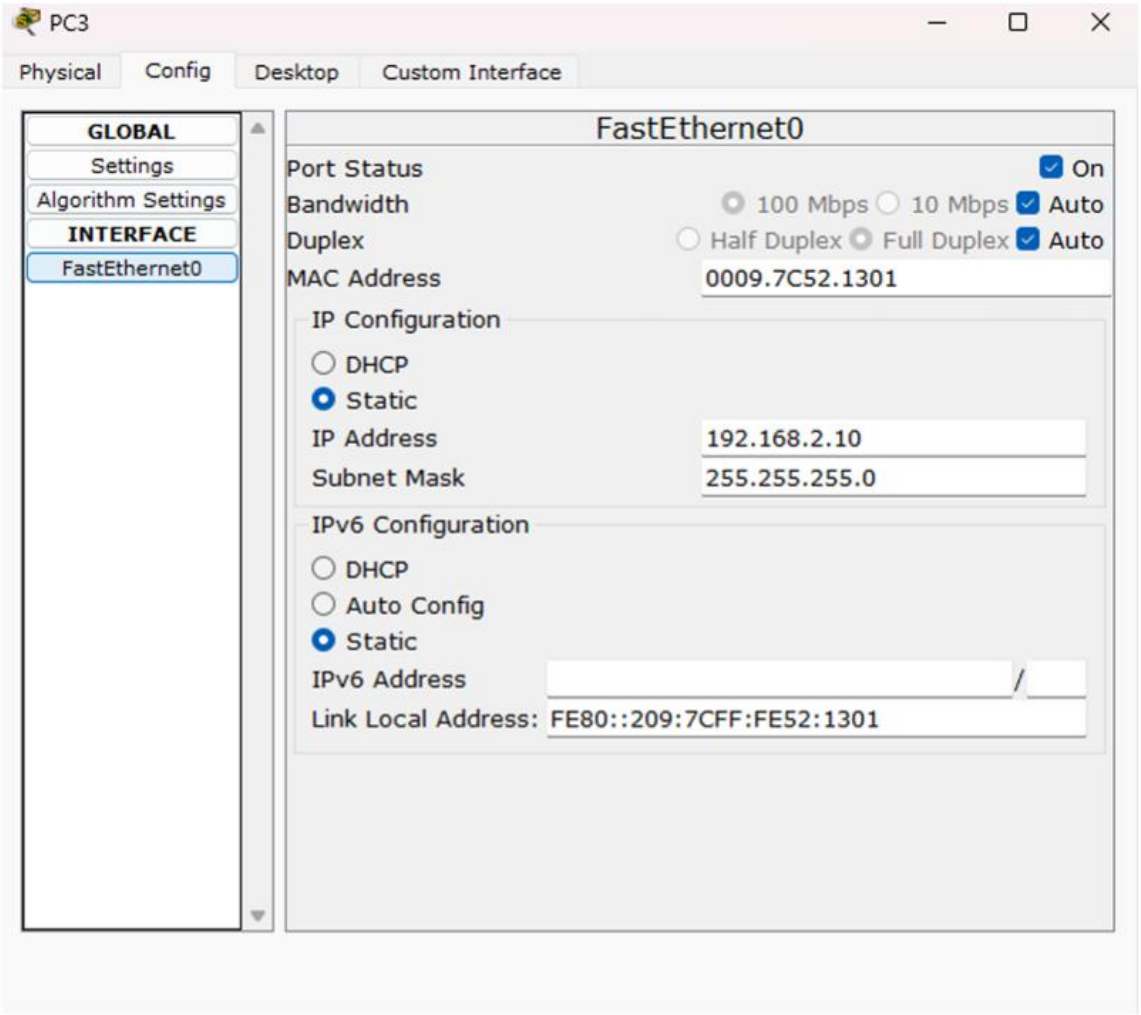


7. Configure PC3

IP Address: 192.168.2.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.

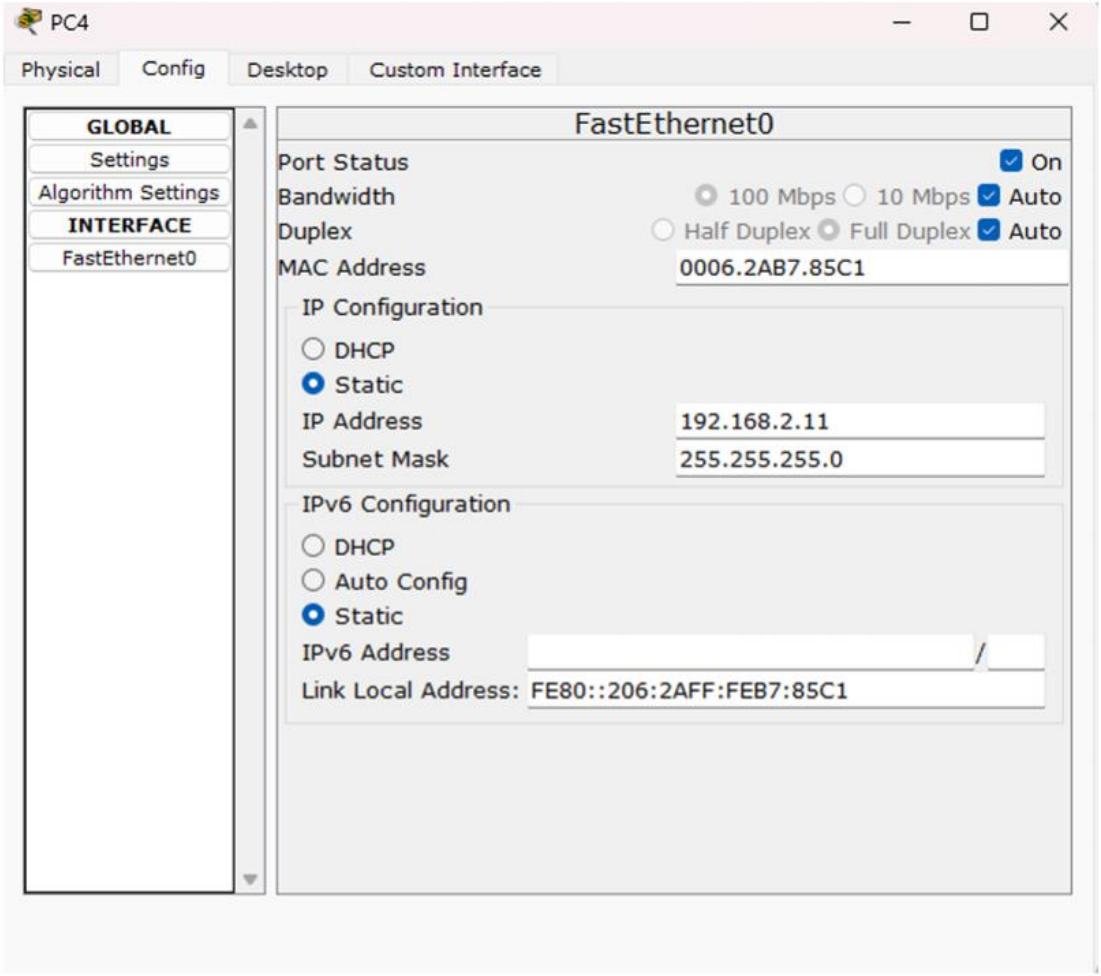


8. Configure PC4

IP Address: 192.168.2.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1



9. Test the network

From **PC1 → Desktop → Command Prompt**, type:

ping 192.168.1.11

This tests PC1 → PC2.

Then:

ping 192.168.2.10

This tests **PC1 → Router → Switch 2 → PC3**.

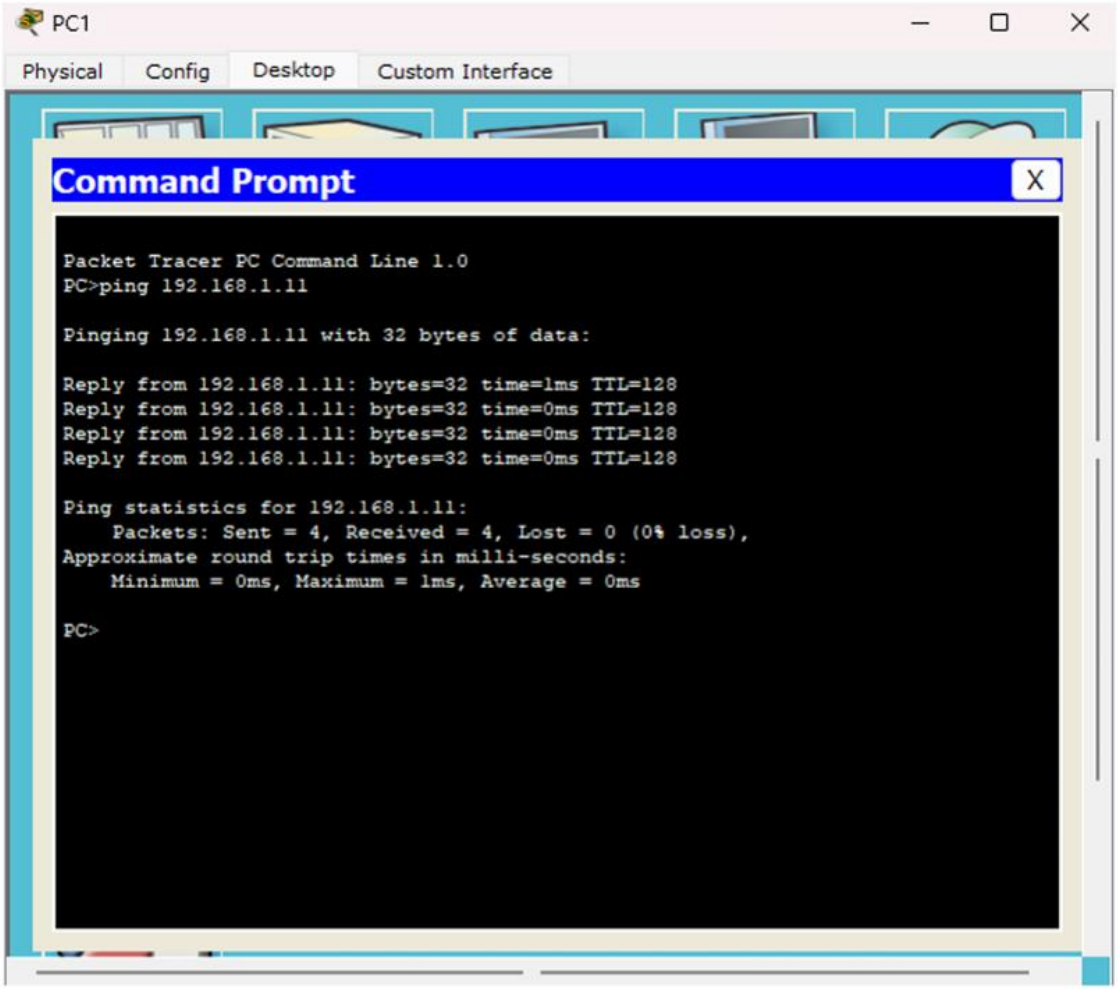
You should receive:

Reply from 192.168.2.10

Reply from 192.168.2.10

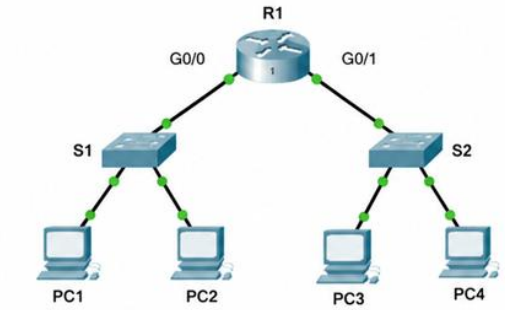
Reply from 192.168.2.10

Reply from 192.168.2.10



SAMPLE CONFIGURATION

TOPOLOGY: One Router, Two Switches and PCs



IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0	192.168.1.1	255.255.255.0	-
R1	G0/1	192.168.2.1	255.255.255.0	-
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1
PC4	NIC	192.168.2.11	255.168.255.0	192.168.2.1

CONFIGURATION STEPS

1. Place Devices

- 1 Router (e.g., 1941)
- 2 Switches (e.g., 2960)
- 4 PCs
- Connections: Copper Straight-Through

2. Configure Router (R1)

```
R1> enable
R1# configure terminal
R1(config)# interface g0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit
R1(config)# interface g0/1
R1(config-if)# ip address 192.168.2.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# end
R1# write memory
```

3. Configure PCs

For PC1

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

For PC2

- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

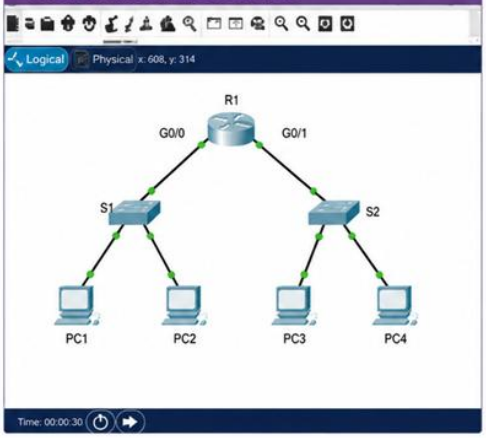
For PC3

- IP Address: 192.168.2.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

For PC4

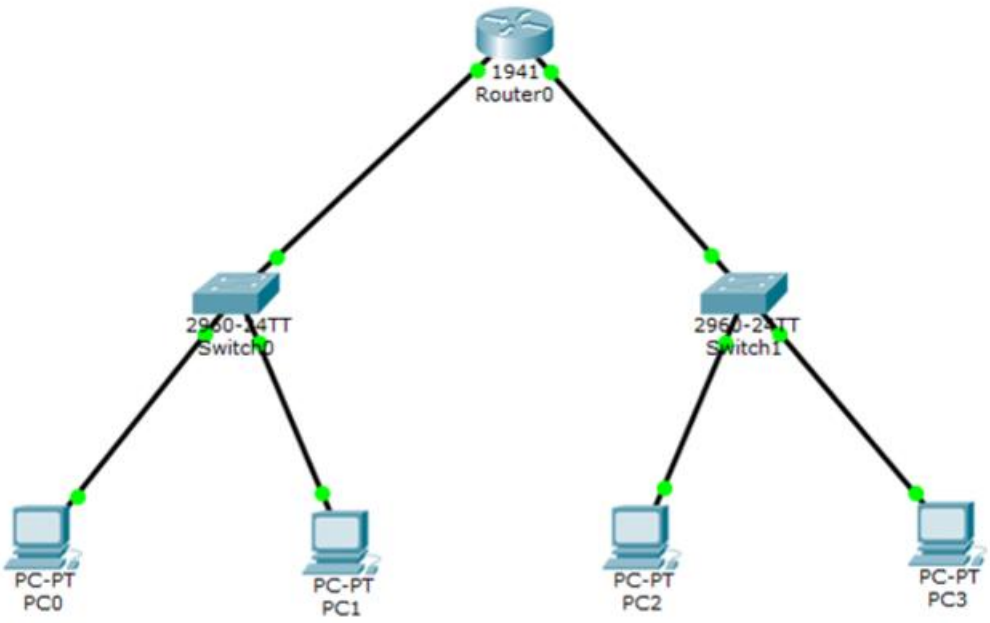
- IP Address: 192.168.2.11
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.2.1

PACKET TRACER TOPOLOGY VIEW



VERIFY CONNECTIVITY

From any PC, open Command Prompt and ping other PCs in the same network and across networks.
Example: From PC1 ping PC3 → `ping 192.168.2.10`
All pings should be successful.



RESULT:

Thus design a simple topology and configure with 1 router, 2 switches, 4 PC using Cisco Packet tracer is verified.